

PART 2: FULL PREDICTED DDCET PAPER

SECTION 1 — Basics of Science and Engineering (Q1–50)

No explanations. Just paper. Like the examiner who enjoys watching panic spread across the hall.

1. Number of significant figures in 0.02050 is:

- (A) 3
 - (B) 4
 - (C) 5
 - (D) 2
-

2. 1 joule is equal to:

- (A) 10^5 erg
 - (B) 10^6 erg
 - (C) 10^7 erg
 - (D) 10^8 erg
-

3. Zero error is an example of:

- (A) Random error
 - (B) Systematic error
 - (C) Personal error
 - (D) Gross error
-

4. Least count of a micrometer screw gauge having pitch 0.5 mm and 50 divisions is:

- (A) 0.1 mm
 - (B) 0.01 mm
 - (C) 0.001 mm
 - (D) 1 mm
-

5. Density of water in CGS system is:

- (A) 1000 g/cm^3
 - (B) 100 g/cm^3
 - (C) 10 g/cm^3
 - (D) 1 g/cm^3
-

6. SI unit of linear momentum is:

- (A) $\text{kg}\cdot\text{m/s}$
 - (B) kg/s
 - (C) $\text{N}\cdot\text{m}$
 - (D) $\text{kg}\cdot\text{m}^2/\text{s}$
-

7. Newton's second law is:

- (A) $F = mv$
 - (B) $F = ma$
 - (C) $P = ma$
 - (D) $P = mv^2$
-

8. A force of 40 N acts on a body of mass 8 kg. Acceleration is:

- (A) 4 m/s^2
 - (B) 5 m/s^2
 - (C) 6 m/s^2
 - (D) 8 m/s^2
-

9. Centripetal force always acts:

- (A) Away from center
 - (B) Towards center
 - (C) Tangentially
 - (D) Vertically upward
-

10. Angular velocity is given by:

- (A) vr
- (B) v/r

(C) r/v

(D) v^2/r

11. Centripetal acceleration is:

(A) v/r

(B) vr

(C) v^2/r

(D) r/v^2

12. Unit of electric flux is:

(A) Nm^2/C

(B) Nm/C

(C) C/Nm

(D) C^2/Nm

13. Number of electrons in 1 coulomb charge is:

(A) 6.25×10^{18}

(B) 6.25×10^{17}

(C) 6.25×10^{19}

(D) 6.25×10^{20}

14. If distance between two charges becomes double, force becomes:

(A) Double

(B) Half

(C) One-fourth

(D) Four times

15. Inverse of resistance is called:

(A) Conductance

(B) Current

(C) Voltage

(D) Resistivity

16. Equivalent resistance of 5Ω , 10Ω and 15Ω in series is:

- (A) 20Ω
 - (B) 25Ω
 - (C) 30Ω
 - (D) 35Ω
-

17. Dielectric inserted between capacitor plates causes capacitance to:

- (A) Increase
 - (B) Decrease
 - (C) Become zero
 - (D) Remain same
-

18. SI unit of capacitance is:

- (A) Volt
 - (B) Coulomb
 - (C) Farad
 - (D) Henry
-

19. Heat capacity per unit mass is called:

- (A) Specific heat
 - (B) Relative heat
 - (C) Thermal conductivity
 - (D) Latent heat
-

20. 37°C in Fahrenheit is:

- (A) 96.8°F
 - (B) 97.6°F
 - (C) 98.6°F
 - (D) 100°F
-

21. When water is boiling, on supplying more heat temperature:

- (A) Increases
- (B) Decreases

- (C) Remains constant
 - (D) Becomes zero
-

22. SI unit of thermal conductivity is:

- (A) J/mK
 - (B) W/mK
 - (C) N/mK
 - (D) K/W
-

23. 104°F equals:

- (A) 38°C
 - (B) 40°C
 - (C) 42°C
 - (D) 36°C
-

24. Audible sound range is:

- (A) 20 Hz to 20 kHz
 - (B) 10 Hz to 10 kHz
 - (C) 20 Hz to 10 kHz
 - (D) 100 Hz to 20 kHz
-

25. Speed of light in vacuum is:

- (A) 3×10^6 m/s
 - (B) 3×10^7 m/s
 - (C) 3×10^8 m/s
 - (D) 3×10^9 m/s
-

26. Light wave is:

- (A) Longitudinal
 - (B) Transverse
 - (C) Mechanical
 - (D) Stationary
-

27. Optical fibre works on principle of:

- (A) Reflection
 - (B) Refraction
 - (C) Total internal reflection
 - (D) Diffraction
-

28. LASER is:

- (A) Divergent
 - (B) Coherent
 - (C) Incoherent
 - (D) Random radiation
-

29. Time period of wave of frequency 50 Hz is:

- (A) 0.2 s
 - (B) 0.02 s
 - (C) 2 s
 - (D) 5 s
-

30. Reverberation time depends on:

- (A) Volume only
 - (B) Absorption only
 - (C) Both volume and absorption
 - (D) Frequency only
-

31. $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$ is:

- (A) Combination reaction
 - (B) Displacement reaction
 - (C) Thermal decomposition reaction
 - (D) Double displacement reaction
-

32. Formula of bleaching powder is:

- (A) CaCO_3
- (B) CaOCl_2

(C) Ca(OH)_2

(D) CaCl_2

33. Milk of magnesia is:

(A) MgO

(B) Mg(OH)_2

(C) MgCl_2

(D) MgSO_4

34. Lustrous non-metal is:

(A) Carbon

(B) Nitrogen

(C) Iodine

(D) Bromine

35. pH of neutral salt solution is:

(A) 5

(B) 6

(C) 7

(D) 8

36. Basic oxide is generally:

(A) Metallic oxide

(B) Non-metallic oxide

(C) Acidic oxide

(D) Neutral oxide

37. Most malleable metal is:

(A) Iron

(B) Gold

(C) Zinc

(D) Copper

38. Rusting of iron is prevented by:

- (A) Painting
 - (B) Greasing
 - (C) Galvanization
 - (D) All of these
-

39. Smallest unit of computer data is:

- (A) Byte
 - (B) Bit
 - (C) KB
 - (D) Nibble
-

40. Ctrl + Home in MS Word moves cursor to:

- (A) End of document
 - (B) Beginning of line
 - (C) Beginning of document
 - (D) End of paragraph
-

41. Correct Excel formula for sum from A1 to A5 is:

- (A) =SUM(A1:A5)
 - (B) =ADD(A1:A5)
 - (C) =TOTAL(A1:A5)
 - (D) =PLUS(A1:A5)
-

42. CPU consists of:

- (A) ALU + CU
 - (B) RAM + ROM
 - (C) Mouse + Keyboard
 - (D) Monitor + CPU
-

43. Which is not an operating system?

- (A) Linux
- (B) UNIX

- (C) MS-DOS
 - (D) MS-Excel
-

44. Relative link in HTML is:

- (A) <http://google.com>
 - (B) ../index.html
 - (C) <https://abc.com>
 - (D) <ftp://site.com>
-

45. Green plants are called:

- (A) Heterotrophs
 - (B) Saprotrophs
 - (C) Autotrophs
 - (D) Parasites
-

46. Bag filters are used to:

- (A) Reduce noise
 - (B) Remove particulate matter
 - (C) Reduce soil pollution
 - (D) Remove smell
-

47. Device converting solar energy directly into electricity:

- (A) Fuel cell
 - (B) Solar heater
 - (C) Photovoltaic cell
 - (D) Boiler
-

48. CFC mainly causes:

- (A) Water pollution
 - (B) Ozone depletion
 - (C) Soil erosion
 - (D) Acid rain
-

49. Paris Agreement aims to keep warming below:

- (A) 1.5°C
 - (B) 5°C
 - (C) 4°C
 - (D) 3.5°C
-

50. PM10 mainly causes:

- (A) Skin problems
 - (B) Respiratory problems
 - (C) Hair fall
 - (D) Vision problems
-

Next:

SECTION 2 (Q51–100)

Mathematics + Soft Skills

Because apparently suffering must come in two sections.

Continue. ..

SECTION 2 — Aptitude Test (Mathematics + Soft Skills)

Q51–100

Now the part where students discover they were “good at maths” only emotionally.

51. If

$$\frac{d}{dx}(e^x + x^2) =$$

- (A) $e^x + 2x$
- (B) $e^x + x$
- (C) xe^x
- (D) $2e^x$

52.

$$\frac{d}{dx}(\log \sin x) =$$

- (A) $\tan x$
 - (B) $\cot x$
 - (C) $-\cot x$
 - (D) $-\tan x$
-

53. If $x + y = 5$, then

$$\frac{dy}{dx} =$$

- (A) 1
 - (B) 0
 - (C) -1
 - (D) 5
-

54.

$$\int e^x dx =$$

- (A) $e^{-x} + C$
 - (B) $e^x + C$
 - (C) $-e^x + C$
 - (D) $xe^x + C$
-

55.

$$\int x^3 dx =$$

- (A) $\frac{x^4}{4} + C$
 - (B) $\frac{x^3}{3} + C$
 - (C) $3x^2 + C$
 - (D) $4x^3 + C$
-

56.

$$\int_0^{\pi} \sin x \, dx =$$

- (A) 0
 - (B) 1
 - (C) 2
 - (D) π
-

57.

$$\log 125 / \log 5 =$$

- (A) 2
 - (B) 3
 - (C) 4
 - (D) 5
-

58.

$$\log 2 + \log 3 =$$

- (A) $\log 5$
 - (B) $\log 6$
 - (C) $2 \log 6$
 - (D) $\log 8$
-

59. Mean of first five negative integers is:

- (A) -2
 - (B) -3
 - (C) -4
 - (D) -5
-

60. Matrix with determinant zero is called:

- (A) Identity matrix
- (B) Singular matrix

- (C) Diagonal matrix
 - (D) Symmetric matrix
-

61.

If

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

then determinant of A is:

- (A) -2
 - (B) 2
 - (C) 10
 - (D) 24
-

62. Inverse of identity matrix is:

- (A) Zero matrix
 - (B) Same identity matrix
 - (C) Diagonal matrix
 - (D) Singular matrix
-

63. 180° in radian is:

- (A) π
 - (B) $\frac{\pi}{2}$
 - (C) 2π
 - (D) $\frac{\pi}{4}$
-

64.

$$\sin 90^\circ =$$

- (A) 0
 - (B) 1
 - (C) -1
 - (D) $\frac{1}{2}$
-

65. Principal period of $\sin 3x$ is:

- (A) 2π
 - (B) $\frac{2\pi}{3}$
 - (C) π
 - (D) $\frac{\pi}{3}$
-

66. Dot product of perpendicular vectors is:

- (A) 1
 - (B) -1
 - (C) 0
 - (D) Infinity
-

67. Unit vector has magnitude:

- (A) 0
 - (B) 1
 - (C) 2
 - (D) Depends on direction
-

68. Slope of line parallel to x-axis is:

- (A) 0
 - (B) 1
 - (C) Undefined
 - (D) Infinite
-

69. Equation of circle with center (0,0) and radius 3 is:

- (A) $x^2 + y^2 = 3$
 - (B) $x^2 + y^2 = 9$
 - (C) $x + y = 3$
 - (D) $x^2 - y^2 = 9$
-

70. If two lines are perpendicular, product of slopes is:

- (A) 1
- (B) 0

(C) -1

(D) 2

71.

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} =$$

(A) 0

(B) 1

(C) Infinity

(D) -1

72.

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n =$$

(A) 1

(B) 2

(C) e

(D) π

73. Adverb in sentence:

"He runs quickly."

(A) He

(B) runs

(C) quickly

(D) runs quickly

74. Select correct option:

"The captain with his players ___ arrived."

(A) have

(B) has

(C) are

(D) were

75. None of the boys ___ present.

- (A) is
 - (B) are
 - (C) was
 - (D) have
-

76. She goes to office ___ bus.

- (A) on
 - (B) in
 - (C) by
 - (D) with
-

77. Select correct spelling:

- (A) Enviroment
 - (B) Environment
 - (C) Enviornment
 - (D) Environmant
-

78. Select correct spelling:

- (A) Entreprenuer
 - (B) Entrepreneur
 - (C) Enterprenuer
 - (D) Enterpreneur
-

79. Choose correct sentence:

- (A) Police has arrested him.
 - (B) Police have arrested him.
 - (C) Police is arrested him.
 - (D) Police arresting him.
-

80. Choose correct sentence:

- (A) I am working here since two years.
 - (B) I have been working here for two years.
 - (C) I working here for two years.
 - (D) I has been working here.
-

81. "Hurrah! We won."

Interjection is:

- (A) We
 - (B) Won
 - (C) Hurrah!
 - (D) Whole sentence
-

82. Communication is a:

- (A) One-way process
 - (B) Two-way process
 - (C) Silent process
 - (D) Written process only
-

83. Body language belongs to:

- (A) Verbal communication
 - (B) Non-verbal communication
 - (C) Written communication
 - (D) Formal communication
-

84. Facial expressions are studied under:

- (A) Kinesics
 - (B) Chronemics
 - (C) Proxemics
 - (D) Grammar
-

85. Stress and anxiety create:

- (A) Physical barrier
 - (B) Psychological barrier
 - (C) Language barrier
 - (D) Mechanical barrier
-

86. Letter of adjustment is usually written in response to:

- (A) Inquiry letter
 - (B) Complaint letter
 - (C) Order letter
 - (D) Offer letter
-

87. A letter asking product details is:

- (A) Complaint letter
 - (B) Order letter
 - (C) Inquiry letter
 - (D) Adjustment letter
-

88. Complimentary close in letter means:

- (A) Opening address
 - (B) Courteous ending
 - (C) Subject line
 - (D) Signature block
-

89. Full form of P.S. in letter writing is:

- (A) Personal Service
 - (B) Post Script
 - (C) Public Service
 - (D) Postal Section
-

90. Receiver's address in business letter is called:

- (A) Heading
- (B) Inside address

(C) Salutation

(D) Subject

Read the passage and answer Q91–95

A library is not merely a building full of books. It is a place where ideas grow, minds expand, and curiosity finds direction. Students who regularly visit libraries develop stronger concentration and better understanding. In the digital age, libraries have also changed by providing access to e-books, research journals, and online learning resources.

91. A library helps mainly in:

- (A) Sleeping peacefully
 - (B) Growing ideas and knowledge
 - (C) Avoiding studies
 - (D) Entertainment only
-

92. Students visiting libraries regularly improve:

- (A) Concentration
 - (B) Laziness
 - (C) Noise level
 - (D) Confusion
-

93. Modern libraries provide:

- (A) Only printed books
 - (B) Only newspapers
 - (C) E-books and journals
 - (D) Only novels
-

94. "Digital age" refers to:

- (A) Stone age
 - (B) Computer and internet era
 - (C) Ancient period
 - (D) Industrial revolution only
-

95. Best title for passage:

- (A) Importance of Libraries
 - (B) Sports Activities
 - (C) Road Safety
 - (D) Pollution Control
-

96. Subject-verb agreement:

"Each of the students ___ present."

- (A) are
 - (B) were
 - (C) is
 - (D) have
-

97. Correct sentence is:

- (A) She do not like tea.
 - (B) She does not likes tea.
 - (C) She does not like tea.
 - (D) She not like tea.
-

98. Choose correct preposition:

"The book is ___ the table."

- (A) in
 - (B) on
 - (C) at
 - (D) with
-

99. Correct spelling is:

- (A) Accomodation
 - (B) Accommodation
 - (C) Acommodation
 - (D) Accommadation
-

100. Select correct sentence:

- (A) Every man and woman were happy.
- (B) Every man and woman was happy.
- (C) Every man and woman are happy.
- (D) Every man and woman have happy.